

Modeling Path Loss in Central Area of Thailand

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1. Introduction

In today's rapidly advancing world, telecommunication systems play a pivotal role in connecting people and enabling access to information. Effective communication infrastructure is crucial, especially in countries like Thailand, where continuous development in this sector supports growing societal and technological needs. Designing and deploying efficient communication systems require consideration of various factors, including geographical features, building density, and user distribution in specific regions.

A key determinant in this process is path loss, which significantly impacts network performance. This research focuses on analyzing path loss using clustering techniques to classify areas based on factors influencing signal propagation, such as 3D gain, building density, and antenna height. These classifications facilitate the development of tailored path loss equations for each group, enhancing the accuracy and efficiency of network planning.

The study aims to provide a systematic approach to addressing the challenges of diverse environmental conditions in communication system design. By refining path loss predictions, this research contributes to improving connectivity, minimizing potential operational issues, and optimizing the overall performance of telecommunication networks. The findings not only address immediate technical needs but also support long-term advancements in communication infrastructure for various regions.

2. Standard Propagation Model

In this research, the Standard Propagation Model (SPM) [1] is utilized to predict path loss. The SPM is specifically designed to provide accurate estimates for signal attenuation in various environments, such as urban, suburban, and rural settings. It accounts for key factors such as frequency, transmitter and receiver heights, and environmental characteristics, offering a more precise approach than generalized models.

The SPM calculates path loss using the following equation (1)

$$LSPM = K_1 + K_2 \log(d) + K_3 \log(HT_{xeff}) + K_4 \text{DiffractionLoss} + K_5 \log(d) \log(HT_{xeff}) + K_6 HR_{xeff} + K_{clutter} f_{clutter} \quad (1)$$

Where:

K_1 : The constant factor in decibels (dB).

K_2 : The multiplier for $\log(d)$

d : The distance between the transmitter and receiver is meters.

K_3 : The multiplier for $\log(HT_{xeff})$

HT_{xeff} : Effective height of the transmitting antenna in meters.

K_4 : The multiplier for *DiffractionLoss*

K_5 : The multiplier for $\log(d) \log(HT_{xeff})$

K_6 : The multiplier for HR_{xeff}

$K_{clutter}$: The multiplier for $f_{clutter}$

$f_{clutter}$: The average clutter loss factor due to other environmental factors.

3. Modeling SPM with clustering technique

The process for developing the path loss model has 4 processes as follows

3.1) Preparing Data

In preparing the data, the process involved two main components for use in the clustering process, as detailed below.

1) Path loss measurement

Path loss can be calculated using Equation (2).

$$\text{Path Loss} = \text{Power} + \text{Antenn 3D Gain} - \text{RSRP} \quad (2)$$

Power and RSRP are values obtained from actual usage data collection, while Antenna 3D Gain can be calculated as follows.

The three-dimensional gain of an antenna (Antenna 3D Gain) can be determined using the interpolation method for three-dimensional directive antenna patterns [2]. This approach is specifically applicable to directional antennas. Denoting the Antenna 3D Gain as $\hat{G}(\theta, \varphi)$, it can be computed as follows.

$$\hat{G}(\theta, \varphi) = \hat{G}_{H_m}(\theta, \varphi) \cdot W_3(\theta, \varphi) + \hat{G}_{V_m}(\theta, \varphi) \cdot [1 - W_3(\theta, \varphi)] \quad (3)$$

Where $\hat{G}_{H_m}(\theta, \varphi)$ and $\hat{G}_{V_m}(\theta, \varphi)$ can be calculated from (4) and (5).

$$\hat{G}_{H_m}(\theta, \varphi) = G_{\theta 1}(\varphi) \cdot W_1(\theta) + G_{\theta 2}(\varphi) \cdot [1 - W_1(\theta)] \quad (4)$$

$$\hat{G}_{V_m}(\theta, \varphi) = G_{\varphi 1}(\theta) \cdot W_2(\varphi) + G_{\varphi 2}(\theta) \cdot [1 - W_2(\varphi)] \quad (5)$$

Where $W_1(\theta)$, $W_2(\theta)$, $W_3(\theta, \varphi)$ can be calculated from (6), (7) and (8).

$$W_1(\theta) = 1 - \frac{2|\theta|}{\pi} \quad (6)$$

$$W_2(\theta) = 1 - \frac{|\varphi|}{\pi} \quad (7)$$

$$W_3(\theta, \varphi) = \frac{2}{\pi} \left[\left| \frac{\pi}{4} - |\theta| + \frac{\pi}{2} - \left| \left(\frac{\pi}{2} - |\varphi| \right) \right| \right| \right] \quad (8)$$

Where can be defined as shown in Figure 1.

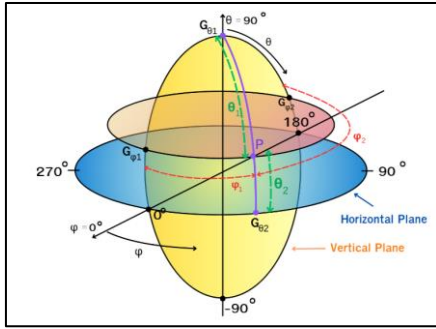


Fig 1. The three-dimensional directive antenna

2) Percentage of building density

The percentage of building density indicates the concentration of buildings within each grid. Since structural density influences path loss, it is determined using Equation (9).

$$\%Building = \frac{\text{Total building area in the grid}}{\text{Grid area}} \quad (9)$$

3.2) Clustering data

The data grouping process utilizes the K-Means clustering technique [4]. We grouped data in 2 rounds.

In first round clustering. Due to the large volume of grid data, we initially perform clustering based on latitude and longitude, grouping nearby grids separately by frequency, with each frequency divided into 500 groups. This preliminary clustering step accelerates the second round of clustering. The example result of first round clustering can be shown in Figure 2.

Next, the clustered data is analyzed to determine the optimal number of clusters for the second round of clustering using the Elbow Method [5] and the number of clusters in each Frequency can show in Table 1.

The second round of clustering is then conducted based on path loss and percentage of building density. This process ensures the creation of appropriately defined clusters, which are subsequently used to develop distinct path loss models tailored to each area. The example result of second round clustering can show in Figure 3 ,4 ,5 and 6.

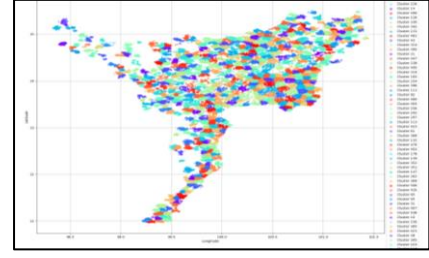


Fig 2. First-round clustering results in 500 groups use Latitude and Longitude at 2100 MHz

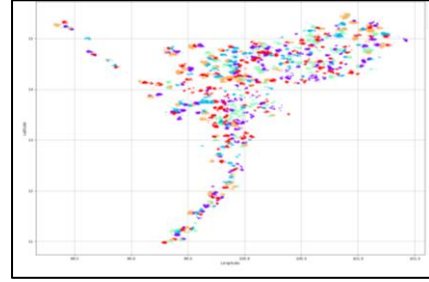


Fig 3. Second-round clustering results in 5 groups using path loss and building density at 700 MHz

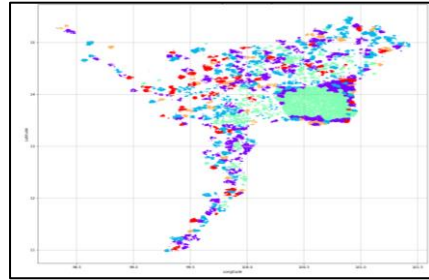


Fig 4. Second-round clustering results in 4 groups using path loss and building density at 900 MHz

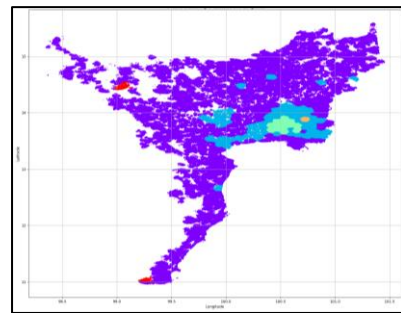


Fig 5. Second-round clustering results in 4 groups using path loss and building density at 1800 MHz

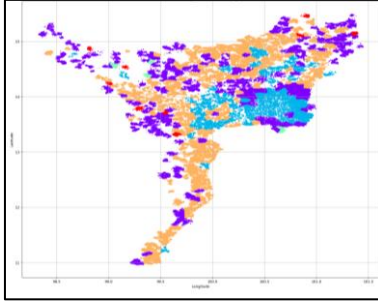


Fig 6. Second-round clustering results in 5 groups using path loss and building density at 2100 MHz

3.3) Curve fitting

In this step, Nonlinear Least Squares [6] is employed to determine the coefficients of the SPM equation (Equation (1)). This method is used to develop models for calculating path loss specific to each clustered area in the central region. We use the number group from clustering in round 2, as shown in Table 1. Additionally, we set the values of *DiffractionLoss* and $f_{(clutter)}$ in the SPM equation to 0, as these parameters could not be derived from the raw data. This approach can be expressed as equation (10), where we need to determine the coefficients K_1, K_2, K_3, K_4 and K_5 to obtain a complete equation

$$LSPM = K_1 + K_2 \log(d) + K_3 \log(HT_{xeff}) + K_4 \log(d) \log(HT_{xeff}) + K_5 HR_{xeff} \quad (10)$$

Table 1. Summary number of clusters in each round

Round	Input	Frequency (MHz)	Number of clusters (Groups)
1	latitude and longitude	700	500
		900	500
		1800	500
		2100	500
2	path loss and percentage of building density	700	5
		900	4
		1800	4
		2100	5

4. Results

In this study, the accuracy of the Standard Propagation Model (SPM) for path loss prediction was evaluated by comparing its results with actual measured path loss data for each frequency. The evaluation involved calculating the RMSE, MSE, and SD for each frequency, as summarized in Table 2, 3, 4 and 5.

Table 2. The error values of the path loss model in 700 MHz

Cluster	RMSE	MSE	SD
1	5.9978	4.3765	2.7653
2	5.2312	3.8600	2.4604
3	5.8616	4.3658	2.8223
4	6.6430	5.0799	5.7503
5	5.6654	4.1707	2.9249

Table 3. The error values of the path loss model in 900 MHz

Cluster	RMSE	MSE	SD
1	8.0501	6.1042	9.0966
2	6.7210	5.0817	8.7827
3	7.4711	5.6688	8.3584
4	8.2906	6.2244	8.4982

Table 4. The error values of the path loss model in 1800 MHz

Cluster	RMSE	MSE	SD
1	12.5247	9.9960	8.7399
2	9.2233	7.0770	12.4757
3	7.4319	5.6250	2.6970
4	7.0905	5.4070	3.0052

Table 5. The error values of the path loss model in 2100 MHz

Cluster	RMSE	MSE	SD
1	12.2139	9.8660	9.2186
2	10.7873	8.3924	10.3422
3	11.5829	9.2413	15.6738
4	11.9310	9.6531	4.7598
5	12.3319	10.0066	7.3017

Additionally, we compared the path loss values obtained from the SPM model with the actual measured path loss values for each frequency. To ensure the relationship between path loss and distance in the SPM equation was more clearly observed, the path loss values obtained from the SPM model were calculated by

fixing the antenna height using the mean of the antenna heights. The results are presented as graphs showing the relationship between path loss and distance, as illustrated in Figures 7, 8, 9, and 10.

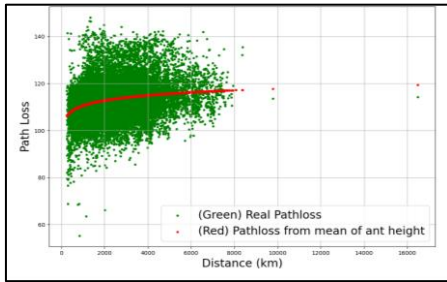


Fig 7. Graphs showing the relationship between path loss and distance at 700 MHz (Cluster 1st)

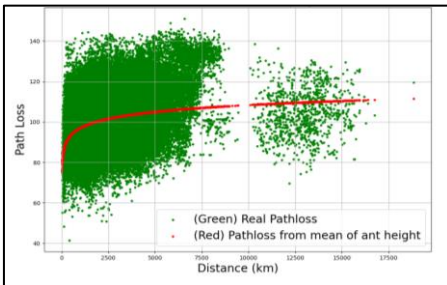


Fig 8. Graphs showing the relationship between path loss and distance at 900 MHz (Cluster 1st)

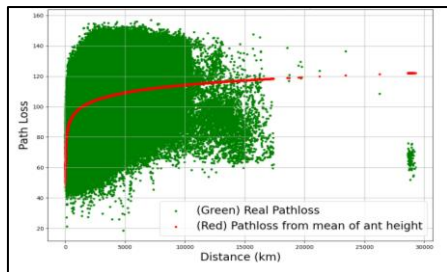


Fig 9. Graphs showing the relationship between path loss and distance at 1800 MHz (Cluster 1st)

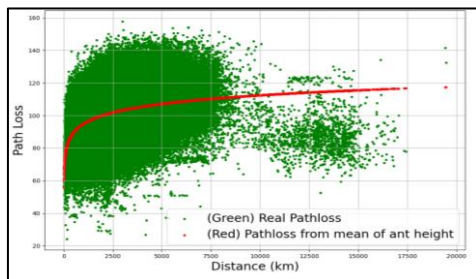


Fig 10. Graphs showing the relationship between path loss and distance at 2100 MHz (Cluster 1st)

5. Conclusions

The study developed a path loss model for central Thailand using clustering based on path loss and building density. A Standard Propagation Model (SPM) was created for each cluster through Nonlinear Least Squares Regression. The model demonstrated acceptable error levels when compared to real-world RSRP and 3D gain measurements, confirming its applicability for path loss prediction and signal quality optimization.

6. Acknowledgement

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7. References

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